



NIMCET 2022

Mathematics:

- If the angle of elevation of the top of a hill from each of the vertices A, B and C of a horizontal triangle is α , then the height of the hill is
(A) $\frac{1}{2}b \tan \alpha \sec B$ (B) $\frac{1}{2}b \tan \alpha \operatorname{cosec} A$
(C) $\frac{1}{2}c \tan \alpha \sin C$ (D) $\frac{1}{2}a \tan \alpha \operatorname{cosec} A$
- If $x^m y^n = (x+y)^{m+n}$, then dy/dx is
(A) $\frac{x+y}{xy}$ (B) xy
(C) x/y (D) y/x
- The value of $3^{3-\log_3 5}$ is
(A) $\frac{5}{27}$ (B) $\frac{27}{5}$
(C) $\frac{9}{5}$ (D) $\frac{5}{9}$
- The value of $\int \frac{(x^2-1)dx}{x^3\sqrt{2x^4-2x^2+1}}$ is
(A) $2\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}+c$ (B) $2\sqrt{2+\frac{2}{x^2}+\frac{1}{x^4}}+c$
(C) $\frac{1}{2}\sqrt{2-\frac{2}{x^2}+\frac{1}{x^4}}+c$ (D) None of the above
- If the volume of a parallelepiped whose adjacent edges are $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} + \alpha\hat{j} + 2\hat{k}$, $\vec{c} = \hat{i} + 2\hat{j} + \alpha\hat{k}$ is 15, then α is equal to
(A) 1 (B) $5/2$
(C) $9/2$ (D) 0
- The area enclosed within the curve $|x| + |y| = 2$ is
(A) 16 sq. unit (B) 24 sq. unit
(C) 32 sq. unit (D) 8 sq. unit
- If $a < b$ then $\int_a^b (|x-a| + |x-b|)dx$ is equal to
(A) $\frac{(b-a)^2}{2}$ (B) $\frac{(b^2-a^2)}{2}$
(C) $\frac{(a^3-b^3)}{2}$ (D) $(b-a)^2$
- If $\hat{a} = \lambda\hat{i} + \hat{j} - 2\hat{k}$, $\hat{b} = \hat{i} + \lambda\hat{j} - 2\hat{k}$, $\hat{c} = \hat{i} + \hat{j} + \hat{k}$, and $[\hat{a} \ \hat{b} \ \hat{c}] = 7$, then the value of λ are
(A) 2, -6 (B) 6, -2
(C) 4, -2 (D) -4, 2
- There are two sets A and B with $|A| = m$ and $|B| = n$. If $|P(A)| - |P(B)| = 112$ then choose the wrong option (where $|A|$ denotes the cardinality of A , and $P(A)$ denotes the power set of A)
(A) $m+n=11$ (B) $2n-m=1$
(C) $2m-n=1$ (D) $3n-m=5$
- A particle is at rest at the origin. It moves along the x -axis with an acceleration $x-x^2$, where x is the distance of the particle at time t . The particle next comes to rest after it has covered a distance
(A) 1 (B) $\frac{1}{2}$
(C) $\frac{3}{2}$ (D) 2
- In a triangle ABC , if the tangent of half the difference of two angles is equal to one third of the tangent of half the sum of the angles, then the ratio of the sides opposite to the angles is
(A) 2 : 1 (B) 1 : 2
(C) 3 : 1 (D) 1 : 1
- The solutions of the equation $4 \cos^2 x + 6 \sin^2 x = 5$ are
(A) $x = n\pi \pm \frac{\pi}{4}$ (B) $x = n\pi \pm \frac{\pi}{3}$
(C) $x = n\pi \pm \frac{\pi}{2}$ (D) $x = n\pi \pm \frac{2\pi}{3}$
- A straight line through the point (4, 5) is such that its intercept between the axes is bisected at A , then its equation is
(A) $3x + 4y = 20$ (B) $3x - 4y + 7 = 0$
(C) $5x - 4y = 40$ (D) $5x + 4y = 40$
- The domain of the function $f(x) = \frac{\cos^{-1}x}{[x]}$ is
(A) $[-1, 0) \cup \{1\}$ (B) $[-1, 1]$
(C) $[-1, 1)$ (D) None of the above
- If the roots of the quadratic equation $x^2 + px + q = 0$ are $\tan 30^\circ$ and $\tan 15^\circ$ respectively, then the value of $2 + p - q$ is
(A) 3 (B) 0
(C) 1 (D) 2





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16. Which of the following is NOT TRUE?

- (A) $\lim_{x \rightarrow \infty} \frac{x}{e^x} = 0$ (B) $\lim_{x \rightarrow 0^+} \frac{1}{xe^x} = 0$
 (C) $\lim_{x \rightarrow 0^+} \frac{\sin x}{1+2x} = 0$ (D) $\lim_{x \rightarrow 0^+} \frac{\cos x}{1+2x} = 0$

17. Inverse of the function $f(x) = \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$ is

- (A) $\log_{10}(2-x)$ (B) $\frac{1}{2} \log_{10} \left(\frac{1+x}{1-x} \right)$
 (C) $\frac{1}{2} \log_{10}(2x-1)$ (D) $\frac{1}{4} \log_{10} \left(\frac{2x}{2-x} \right)$

18. The first three moments of a distribution about 2 are 1, 16, -40 respectively. Then mean and variance of the distribution are

- (A) (2, 16) (B) (2, 15)
 (C) (3, 15) (D) (1, 16)

19. If $a_1, a_2, \dots, a_n$ are in Arithmetic Progression with common difference, d , then the sum $(\sin d)(\operatorname{cosec} a_1 \cdot \operatorname{cosec} a_2 + \operatorname{cosec} a_2 \operatorname{cosec} a_3 + \dots + \operatorname{cosec} a_{n-1} \operatorname{cosec} a_n)$ is equal to

- (A) $\cot a_1 - \cot a_n$ (B) $\sin a_1 - \sin a_n$
 (C) $\operatorname{cosec} a_1 - \operatorname{cosec} a_n$ (D) $a_1 - a_n$

20. Solutions of the equation $\tan^{-1} \sqrt{x^2 + x} + \sin^{-1} \sqrt{x^2 + x + 1} = \frac{\pi}{2}$ are

- (A) 0, 1 (B) 1, -1
 (C) 0, -1 (D) 0, -2

21. If $\operatorname{cosec} \theta - \cot \theta = 2$, then the value of $\operatorname{cosec} \theta$ is

- (A) $\frac{5}{3}$ (B) $\frac{3}{5}$
 (C) $\frac{4}{5}$ (D) $\frac{5}{4}$

22. The function $f(x) = \begin{cases} (1+2x)^{\frac{1}{x}}, & x \neq 0 \\ e^2, & x = 0 \end{cases}$ is

- (A) Differentiable at $x = 0$ (B) Continuous at $x = 0$
 (C) Discontinuous at $x = 0$ (D) Not differentiable at $x = 0$

23. If the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ are coincide, then the value of b^2 is

- (A) 25 (B) 16
 (C) 64 (D) 49

24. If $(\hat{a} \times \hat{b}) \times \hat{c} = \hat{a} \times (\hat{b} \times \hat{c})$, then

- (A) $\hat{a}$ and $\hat{b}$ are collinear
 (B) $\hat{a}$ and $\hat{b}$ are perpendicular
 (C) $\hat{a}$ and $\hat{c}$ are collinear
 (D) $\hat{a}$ and $\hat{c}$ are perpendicular

25. Suppose that the temperature at a point (x, y) on a metal plate is $T(x, y) = 4x^2 - 4xy + y^2$. An ant, walking on the plate, traverses a circle of radius 5 centered at the origin. What is the highest temperature encountered by the ant?

- (A) 125 (B) 120
 (C) 0 (D) 25

26. The function $f(x) = \log(x + \sqrt{x^2 + 1})$ is

- (A) an even function
 (B) an odd function
 (C) a periodic function
 (D) neither an even nor an odd function

27. Angles of elevation of the top of a tower from three points (collinear) A, B and C on a road leading to the foot of the tower are $30^\circ, 45^\circ$ and 60° respectively. The ratio of AB and BC is

- (A) $\sqrt{3} : 1$ (B) $\sqrt{3} : 2$
 (C) $1 : 2$ (D) $2 : \sqrt{3}$

28. The mean of 25 observations was found to be 38. It was later discovered that 23 and 38 were misread as 25 and 36, then the mean is

- (A) 32 (B) 36
 (C) 38 (D) 42

29. In a Harmonic Progression, p^{th} term is q and q^{th} term is p . Then pq^{th} term is

- (A) 0 (B) 1
 (C) pq (D) $pq(p+q)$

30. A four-digit number is formed using the digits 1, 2, 3, 4, 5 without repetition. The probability that it is divisible by 3 is

- (A) $\frac{1}{3}$ (B) $\frac{1}{4}$
 (C) $\frac{1}{5}$ (D) $\frac{1}{6}$

31. Let $\hat{a} = 2\mathbf{i} + 2\mathbf{j} + \mathbf{k}$ and $\hat{b}$ be another vector such that $\hat{a} \cdot \hat{b} = 14$ and $\hat{a} \times \hat{b} = 3\mathbf{i} + \mathbf{j} - 8\mathbf{k}$ the vector $\hat{b} =$





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(A) $5i + j + 2k$

(B) $5i - j - 2k$

(C) $5i + j - 2k$

(D) $3i + j + 4k$

32. A survey is done among a population of 200 people who like either tea or coffee. It is found that 60% of the population like tea and 72% of the population like coffee. Let x be the number of people who like both tea & coffee. Let $m \leq x \leq n$, then choose the correct option.

(A) $n - m = 56$

(B) $n - m = 28$

(C) $n - m = 32$

(D) $n + m = 92$

33. The eccentricity of an ellipse, with its center at the origin is $\frac{1}{3}$. If one of the directrices is $x = 9$, then the equation of ellipse is

(A) $9x^2 + 8y^2 = 72$

(B) $8x^2 + 9y^2 = 72$

(C) $8x^2 + 7y^2 = 56$

(D) $7x^2 + 8y^2 = 56$

34. For $\alpha \in \mathbb{R}$ (the set of all real numbers), $\alpha \neq -1$,

$$\lim_{n \rightarrow \infty} \frac{(1^\alpha + 2^\alpha + \dots + n^\alpha)}{(n+1)^{\alpha-1}[(n\alpha+1) + (n\alpha+2) + \dots + (n\alpha+n)]} = \frac{1}{60}. \text{ Then one of the values of } \alpha \text{ is}$$

(A) 5

(B) 8

(C) $-15/2$

(D) $-17/2$

35. Let a be the distance between the lines $-2x + y = 2$ and $2x - y = 2$, and b be the distance between the lines $4x - 3y = 5$ and $6y - 8x = 1$, then

(A) $40b = 11\sqrt{5}a$

(B) $40\sqrt{2}a = 11b$

(C) $11\sqrt{2}b = 40a$

(D) $11\sqrt{2}a = 40b$

36. If $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2+x & 1 \\ 1 & 1 & 2+y \end{vmatrix}$ for $x \neq 0, y \neq 0$, then D

is

(A) Divisible by x and y

(B) Divisible by x but not by y

(C) Divisible by $(1+x)$ and $(1+y)$

(D) Divisible by $(1+x)$ but not $(1+y)$

37. If $0 < P(A) < 1$ and $0 < P(B) < 1$, and $P(A \cap B) = P(A)P(B)$, then

(A) $P(B|A) = P(B) - P(A)$

(B) $P(A^c - B^c) = P(A^c) - P(B^c)$

(C) $P(A \cup B)^c = P(A^c)P(B^c)$

(D) $P(A|B) = P(A) - P(B)$

38. let a, b, c be distinct non-negative numbers. If the vectors $a\hat{i} + a\hat{j} + c\hat{k}$, $\hat{i} + \hat{k}$ and $c\hat{i} + c\hat{j} + b\hat{k}$ lie in a plane, then c is

(A) The Arithmetic Mean of a and b

(B) The Geometric Mean of a and b

(C) The Harmonic Mean of a and b

(D) Equal to zero

39. The value of $\cot\left(\operatorname{cosec}^{-1}\frac{5}{3} + \tan^{-1}\frac{2}{3}\right)$ is

(A) $6/17$

(B) $3/17$

(C) $4/17$

(D) $5/17$

40. Which term of the series $\frac{\sqrt{5}}{3}, \frac{\sqrt{5}}{4}, \frac{1}{\sqrt{5}}, \dots$ is $\frac{\sqrt{5}}{13}$?

(A) 12

(B) 11

(C) 10

(D) 9

41. Coordinate of focus of the parabola $4y^2 + 12x - 20y + 67 = 0$ is

(A) $\left(-\frac{5}{4}, \frac{17}{2}\right)$

(B) $\left(-\frac{17}{2}, \frac{5}{4}\right)$

(C) $\left(-\frac{17}{4}, \frac{5}{2}\right)$

(D) $\left(-\frac{5}{2}, \frac{17}{4}\right)$

42. The 10th and 50th percentiles of the observations 32, 49, 23, 29, 118 respectively are

(A) 21, 32

(B) 23, 32

(C) 23, 33

(D) 22, 31

43. Area of the parallelogram formed by the lines $y = 4x$, $y = 4x + 1$, $x + y = 0$ and $x + y = 1$ is

(A) $1/5$

(B) $2/5$

(C) 5

(D) 10

44. If $a_1, a_2, \dots, a_n$ are any real numbers and n is any positive integer, then

(A) $n \sum_{i=1}^n a_i^2 < (\sum_{i=1}^n a_i)^2$

(B) $n \sum_{i=1}^n a_i^2 \geq (\sum_{i=1}^n a_i)^2$

(C) $\sum_{i=1}^n a_i^2 \geq (\sum_{i=1}^n a_i)^2$

(D) None of the above

45. There are two circles in xy -plane whose equations are $x^2 + y^2 - 2y = 0$ and $x^2 + y^2 - 2y - 3 = 0$. A point (x, y) is chosen at random inside the larger circle. Then the probability that the point has been taken from smaller circle is

(A) $1/3$

(B) $2/3$

(C) $1/2$

(D) $1/4$



46. $f(x) = x + |x|$ is continuous for

- (A) $x \in (-\infty, \infty)$ (B) $x \in (-\infty, \infty) - \{0\}$
(C) only $x > 0$ (D) No value of x

47. The correct expression for $\cos^{-1}(-x)$ is

- (A) $\frac{\pi}{2} - \cos^{-1}x$ (B) $\pi - \cos^{-1}x$
(C) $\pi + \cos^{-1}x$ (D) $\frac{\pi}{2} + \cos^{-1}x$

48. If $\cos^{-1}\frac{x}{2} + \cos^{-1}\frac{y}{3} = \Phi$, then $9x^2 - 12xy\cos\Phi + 4y^2$ is equal to

- (A) $-36\sin^2\Phi$ (B) $36\sin^2\Phi$
(C) $36\cos^2\Phi$ (D) 36

49. If $\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$, ($a > b$) and $x^2 - y^2 = c^2$ cut at right angles, then

- (A) $a^2 + b^2 = 2c^2$ (B) $b^2 - a^2 = 2c^2$
(C) $a^2 - b^2 = 2c^2$ (D) $a^2 - b^2 = c^2$

50. If α, β are the roots of $x^2 - x - 1 = 0$, and $A_n = \alpha^n + \beta^n$, then Arithmetic Mean of A_{n-1} and A_n is

- (A) $2A_{n-1}$ (B) $\frac{1}{2}A_{n+1}$
(C) $2A_n - 2$ (D) None of the above

Reasoning/Aptitude:

51. U, V, W, X and T are sitting on a bench. T is sitting next to U, V is sitting next to W, W is not sitting with X who is on the left end of the bench. V is in the second position from the right. T is to the right of U and X. T and V are sitting together. In which position T is sitting?

- (A) Between U and W (B) Between V and X
(C) Between X and W (D) Between U and V

52. DNN, FQQ, HTT, _____, LZZ

- (A) JWW (B) JVV
(C) JXX (D) IWW

53. Find the synonym that is most nearly similar in meaning to the word:

DEBACLE

- (A) Catastrophe (B) Corker
(C) Opulence (D) Dandy

54. Fill in the blank: HEC, JGE, LIG, NKI, _____

- (A) ONM (B) PMK
(C) KMP (D) HGF

55. Replace the question mark with an appropriate image to complete the analogous pair.



(A)



(B)



(C)



(D)

(A)

C

(C) D

(B) B

(D) A

56. Which of the following is the odd one from the given alternatives?

- (A) Driving (B) Sailing
(C) Diving (D) Swimming

57. Select the pair of words, which are related in the same way as the capitalized words are related to each other.

Frugal : Extravagant

- (A) Predecessor : Precursor (B) Hermit : Philosopher
(C) Teacher : Philanthropist (D) Criticise : Advocate

58. Identify the sixth number in the series:

6, 11, 21, 36, 56, ?

- (A) 52 (B) 81
(C) 82 (D) 21

59. Looking at the portrait of a man, Manu said, "Her mother is the wife of my father's son and I have no brother and sister." Whose portrait was Manu looking at?

- (A) His son (B) His daughter
(C) His father (D) His nephew

60. Select the pair of words, which are related in the same way as the capitalized words are related to each other.

BUTTERFLY : FREEDOM

- (A) Chicken : Rooster
- (B) Self-reliant : Buoyant
- (C) Frog : Water
- (D) Horse : Speed

61. If in a certain language, KOLKOTA is coded as LPMLPUB, how is MUMBAI coded in that code?

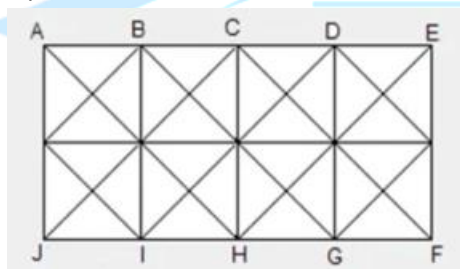
- (A) NVNBCJ
- (B) OVNBH
- (C) NVNCBJ
- (D) NUNBCH

62. Select the related word from the given alternatives.

MELT: LIQUID :: FREEZE: ?

- (A) PUSH
- (B) CONDENSE
- (C) SOLID
- (D) ICE

63. In the following figure, find the total number of squares.

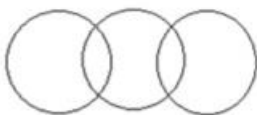


- (A) 20
- (B) 36
- (C) 18
- (D) 24

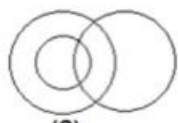
64. Which of the following diagram correctly represents lions, elephants, and animals?



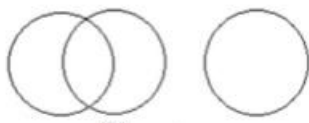
(A)



(B)



(C)



(D)

- (A) C
- (B) A
- (C) B
- (D) D

65. Identify the fifth number in the series:
122, 144, 166, 188, ?

- (A) 210
- (B) 234
- (C) 345
- (D) 310

66. Fill in the blank: JAK, KBL, LCM, MDN, _____

- (A) MEN
- (B) NEO
- (C) OEP
- (D) PEQ

67. Deepak, Rahul, Manoj, and Vinod are brothers. Who is the heaviest?

I. Rahul is heavier than Deepak and Vinod, but lighter than Manoj.

II. Deepak is lighter than Rahul and Manoj, but heavier than Vinod.

- (A) Either I or II is sufficient
- (B) Data in both the statements together are not sufficient
- (C) Statement I alone is sufficient, but statement II alone is not sufficient
- (D) Statement II alone is sufficient, but statement I alone is not sufficient

68. Today is Wednesday. What would be the day after 61 days?

- (A) Monday
- (B) Saturday
- (C) Sunday
- (D) Tuesday

69. Choose the word opposite in meaning to the given word: MITIGATE

- (A) Tranquelize
- (B) Intensify
- (C) Alleviate
- (D) Abate

Directions [Q. Nos. 70 to 72]: In each question below are given two statements followed by two conclusions numbered I and II. You have to take the given two statements to be true even if they seem to be at variance from commonly known facts.

Read the conclusion and then decide which of the given conclusions logically follows from the two given statements, disregarding commonly known facts.

70. Statements: No women teacher can play. Some women teachers are athletes

Conclusions:

- I. Make athletes can play
- II. Some athletes can play

- (A) Only Conclusion I follows
- (B) Either I or II follows



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- (C) Only conclusion II follows
(D) Neither I nor II follows

71. Statements: All mangoes are golden in colour.
No golden-coloured things are cheap.

Conclusions:

- I. All mangoes are cheap.
II. Golden-coloured mangoes are not cheap.
(A) Only Conclusion II follows
(B) Only conclusion I follows
(C) Either I or II follows
(D) Neither I nor II follows

72. Statements: All young scientists are open-minded.
No open-minded men are superstitious

Conclusions:

- I. No scientist is superstitious.
II. No young people are superstitious.
(A) Neither I nor II follows
(B) Only conclusion II follows
(C) Only Conclusions I follows
(D) Either I or II follows

73. Find out the wrong number in the FOLLOWING series.

2, 5, 10, 17, 26, 38, 50, 65

- (A) 65 (B) 38
(C) 26 (D) 50

74. Find out the wrong number in the FOLLOWING series.

30, -5, -45, -90, -145, -195, -255

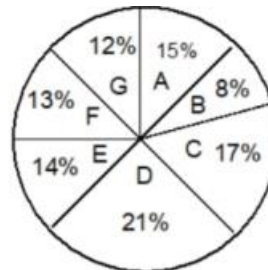
- (A) -195 (B) -5
(C) -145 (D) -255

75. Six books are labelled A, B, C, D, E, and F and are placed side by side. Books B, C, E and F have green covers while others have yellow covers. Books A, B and D are new while the rest are old volumes. Books A, B and C are law reports while the rest are medical extracts. Which two books are old medical extracts and have green covers?

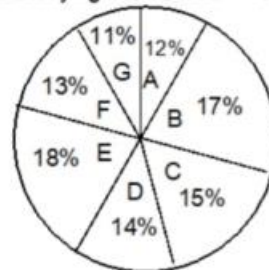
- (A) C and F (B) C and E
(C) B and C (D) E and F

Directions [Q. Nos. 76 to 78]: The following questions are based on the pie-charts given below. Percentage-

wise distribution of student studying in Arts and Commerce in seven different institutions
Different institutes – A, B, C, D, E, F and G
Total number of students studying Arts = 3800



Total number of students studying Commerce = 4200



76. What is the total number of students studying Arts in Institutes A and G together?

- (A) 1206 (B) 1126
(C) 1026 (D) 1226

77. How many students from institute B study Arts and Commerce?

- (A) 1180 (B) 1208
(C) 1018 (D) 1108

78. The ratio of the number of students studying Arts to that studying Commerce in Institute E is

- (A) 19 : 27 (B) 19 : 16
(C) 19 : 28 (D) 27 : 14

79. Statement: Many shops in the local market have extended their shops and occupied most part of the footpath in front of their shops.
Course of Action:

- I. The civic authority should immediately activate a task force to clear all the footpaths encroached by the shop owners.
II. The civic authority should charge hefty penalty to the shop owners for occupying the footpath.
III. The civic authority should setup a monitoring system so that encroachments do not recur in future.





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- (A) I and II follow
(C) II and III follow

- (B) All I, II and III follows
(D) None follows

80. Statement: There is a significant increase in the number of patients affected by some disease in a city.
Course of action:

I. Municipal Corporation of the city should take immediate action.

II. This problem should be raised in the UNESCO.

III. Hospitals in the city should be equipped properly for the treatment of the patients.

- (A) I and III follow
(C) All follow
- (B) I and II follow
(D) Only III follow

Direction[Q. Nos. 81 to 85]: read the following information carefully and answer the questions.
Five Dramas A, B, C, D and E have to be staged in 6 hour where 1 hour needs to be given per drama.

- 1) A break of 1 hour has to be taken in third or four hour.
- 2) Drama show cannot be started with A and cannot end in C.
- 3) D has to follow B immediately with no break in between.
- 4) A cannot be done immediately after D
- 5) A has to precede E immediately with no break in between.

81. Which hour is a break hour?

- (A) 5th
(C) 3rd
- (B) 2nd
(D) 4th

82. Which is the drama to be staged first?

- (A) B
(C) D
- (B) None of these
(D) A

83. Which is the drama staged immediately after the break?

- (A) A
(C) D
- (B) B
(D) None of these

84. Which drama is staged immediately after D?

- (A) E
(C) C
- (B) None of these
(D) B

85. Which drama is staged immediately after E?

- (A) A
(C) None of these
- (B) E
(D) C

86. The greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5 respectively is

- (A) 127
(C) 305
- (B) 235
(D) 123

87. At what time, in minutes, between 3 o'clock and 4 o'clock, both the needles will coincide each other?

- (A) $5\frac{1}{11}$
(C) $13\frac{4}{11}$
- (B) $12\frac{4}{11}$
(D) $16\frac{4}{11}$

88. 1. A is the brother of B.

2. C is the father of A.

3. D is brother of E.

4. E is the daughter of B.

Then, the uncle of D is?

- (A) E
(C) B
- (B) A
(D) C

89. Running at the same constant rate, 6 identical machines can produce a total of 270 bottles per minute. At this rate, how many bottles could 10 such machines produce in 4 minutes?

- (A) 648
(C) 10800
- (B) 2700
(D) 1800

90. A person's present age is two-fifth of the age of his mother. After 8 years, he will be one-half of the age of his mother. What is the present age of the mother?

- (A) 40
(C) 50
- (B) 30
(D) 60





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Computer:

91. Suppose the largest n bit number requires ' d ' digits in decimal representation. Which of the following relations between ' n ' and ' d ' is approximately correct
- (A) $d = 2^n$ (B) $n = 2^d$
(C) $d < n \log_{10} 2$ (D) $d > n \log_{10} 2$
92. 'Floating point representation' is used to represent
- (A) Boolean Values (B) Integers
(C) Whole Numbers (D) Real Numbers
93. Which of the following is equivalent to the Boolean expression:
 $(X + Y) \cdot (X + \bar{Y}) \cdot (\bar{X} + Y)$
- (A) XY (B) $X\bar{Y}$
(C) $\bar{X}Y$ (D) $\bar{X}\bar{Y}$
94. The Boolean expression $AB + AB' + A'C + AC$ is unaffected by the value of the Boolean variable
- (A) B (B) None of these
(C) A (D) C
95. Write the simplified form of the Boolean expression $(A + C)(AD + AD') + AC + C$:
- (A) $A + C$ (B) $A' + C$
(C) $A + C'$ (D) $A + D$
96. If a processor clock is rated as 2500 million cycles per second, then its clock period is:
- (A) 2.50×10^{-10} sec (B) 4.00×10^{-10} sec
(C) 1.00×10^{-10} sec (D) None of the above
97. The minimum number of NAND gates required for implementing the Boolean expression, $AB + A\bar{B}C + A\bar{B}\bar{C}$ is:
- (A) 1 (B) 0
(C) 2 (D) 3
98. FFFF will be the last memory location in a memory of size
- (A) 1k (B) 16k
(C) 32k (D) 64k
99. The maximum and minimum value represented in signed 16 bit 2's complement representations are
- (A) 0 and 65535 (B) - 32768 and 32767
(C) -16384 and 16383 (D) 0 and 32767
100. If a signal passing through a gate is inhibited by sending a low into one of the inputs, and the output is HIGH, the gate is a(n) :
- (A) NOR (B) AND
(C) NAND (D) OR

English:

101. Fill in the blanks with the correct option
Technical writing demands ____ use of language
- (A) Dramatic (B) Factual
(C) Figurative (D) Poetic
102. Select the correct form of verb/Subject verb agreement:
The principal, along with his assistants, _____ the meeting.
- (A) attend (B) are attending
(C) attending (D) is attending
103. Select correct articles
He is M.A. with PhD and teaches in university.
- (A) a, an (B) a, the
(C) the, the (D) an, the
104. Choose the correct alternative
The country cleared this path and paved it with packed gravel, so they would have a peaceful place to hike and bike.
Which of the following alternatives to the underlined portion would NOT be acceptable?
- (A) path, paving (B) Path and then paved
(C) path before paving (D) path paved
105. "Bite and bullet" means
- (A) to analyze your faults
(B) to accept something that is difficult or unpleasant
(C) to stop a conflict
(D) to become mad





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106. What can you call a person who leads an unconventional style of living?

- (A) Cynic (B) Agnostic
(C) Bohemian (D) Altruist

107. Fill in the blank with the most appropriate option
Kedar _____ this project for a month and now he is about to join a new project.

- (A) guiding (B) has been guiding
(C) guided (D) guides

Directions [Q. Nos. 108 and 109]: Read the following passage carefully and answer the questions:
You might think you've experienced VR, and you might have been pretty impressed. Particularly, if you're a gamer, there are some great experiences to be had out there (or rather, in there) today. But over the next few years, in VR, as in all fields of technology, we're going to see things that make what is cutting-edge today look like Space invaders. And although the games will be amazing, the effects of this transformation will be far broader, touching on our work, education, and social lives. Today's most popular VR applications involve taking total control of a user's senses (sight and hearing, particularly) to create a totally immersive experience that places the user in a fully virtual environment that feels pretty realistic. Climb up something high and look down, and you're likely to get a sense of vertigo. If you see an object moving quickly towards your head, you'll feel an urge to duck out of the way.

Very soon, VR creators will extend this sensory hijacking to our other faculties – for example, touch and smell – to deepen that sense of immersion. At the same time, the devices we use to visit these virtual worlds will become cheaper and lighter, removing the friction that can currently be a barrier.

I believe extended reality (XR) – a team that covers virtual reality (VR), augmented reality (AR), and mixed reality (MR) – will be one of the most transformative tech trends of the next five years. It will be enabled and augmented by other tech trends, including super-fast networking, that will let us experience VR as a cloud service just like we currently consume music and movies. And artificial intelligence (AI) will provide us with more

personalized virtual worlds to explore, even giving us realistic virtual characters to share our experience with.

108. The passage states all the following about VR applications except

- (A) Vertigo is a major feature of all AI application
(B) VR applications takes control of the user's senses
(C) VR applications creates a virtual environment that feels pretty realistic
(D) Future AI will allow us to share our experiences with realistic virtual characters

109. Select an antonym for the word 'augment' from the options given below:

- (A) Aggrandize (B) Reinforce
(C) Inflate (D) Curtail

110. 'Duck out of something' means

- (A) To fall down
(B) To meet with an accident
(C) To avoid doing something
(D) To hit something hard

111. Which of the phrases given below should replace the phrase printed in bold to make the sentence grammatically correct.

He is addicted to smoke.

- (A) used to smoke (B) addicted to smoking
(C) addicted with smoking (D) addict of smoking

112. Select the one which best expresses the same sentence in indirect/direct speech:

He said, "I am glad to be here this evening"

- (A) He said that he was glad to be there that evening
(B) He says he was glad to be here this evening.
(C) He asked he is glad to be here this evening
(D) He said he was glad to be here this evening

113. Choose the synonym.

LIBERAL

- (A) Reactionary (B) Generous
(C) Sober (D) Affectionate

114. Identify the word that is similar in meaning to the underlined word. Raghu made adulatory remarks about the waiter who served the food.





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- (A) Derogatory (B) Ironic
(C) Slanderous (D) Complimentary

115. Select the word which means the same as the following:

To read something carefully

- (A) Presume (B) Perverse
(C) Persiflage (D) Peruse

116. Choose the antonym:

BOLD

- (A) Nervous (B) Coy
(C) Timid (D) Fearful

117. Fill in the blanks with the correct preposition
Jagdish is waiting for me _____ the campus.

- (A) On (B) At
(C) In (D) Out

118. Choose the correct alternative with the correct choice given below each statement:

You are to confirm _____ the rules of the institute.

- (A) to (B) with
(C) of (D) on

119. Read the following sentence to find if there is any error in any part:

(a) If I were him/ (b) I would teach/ (c) him a lesson/ (d)

No error

- (A) No error (B) If I were him
(C) I would teach (D) him a lesson

120. Fill in the blank

Neither Peter nor I _____ responsible for this blunder.

- (A) am (B) are
(C) is (D) were





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Answer Key:

1. D	13. D	25. A	37. C	49. C	61. C	73. B	85. C	97. B	109.C
2. D	14. A	26. B	38. B	50. B	62. C	74. C	86. A	98. D	110.D
3. B	15. C	27. A	39. A	51. D	63. D	75. D	87. D	99. B	111.B
4. C	16. D	28. C	40. B	52. A	64. B	76. C	88. B	100.C	112.A
5. C	17. B	29. B	41. C	53. A	65. A	77. C	89. D	101.B	113.B
6. D	18. C	30. C	42. B	54. B	66. B	78. A	90. A	102.D	114.D
7. D	19. A	31. A	43. A	55. C	67. C	79. B	91. C	103.D	115.D
8. A	20. C	32. A	44. B	56. A	68. A	80. A	92. D	104.D	116.C
9. C	21. D	33. B	45. D	57. D	69. B	81. C	93. A	105.B	117.A
10. C	22. A	34. B, D	46. A	58. B	70. D	82. A	94. A	106.C	118.A
11. A	23. B	35. A	47. B	59. A	71. A	83. D	95. A	107.B	119.A
12. A	24. C	36. C	48. B	60. D	72. A	84. C	96. B	108.A	120.A

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Solution

Mathematics:

1. The correct option is (D) $\frac{1}{2} \tan \alpha \operatorname{cosec} \alpha$

Clearly, the foot of the tower is the circum-centre of $\triangle ABC$. Let R be the circum-radius of $\triangle ABC$.

Then,

$$R = h \cot \alpha$$

$$\Rightarrow \frac{a}{2 \sin A} = h \cot \alpha$$

$$\Rightarrow h = \frac{a}{2} \tan \alpha \operatorname{cosec} \alpha$$

2. The correct option is (D) $\frac{y}{x}$

$$x^m y^n = (x+y)^{m+n}$$

Taking log both sides, we get

$$\Rightarrow m \log x + n \log y = (m+n) \log (x+y)$$

Differentiating both sides w.r.t. x, we get

$$\Rightarrow \frac{m}{x} + \frac{n}{y} \frac{dy}{dx} = \frac{(m+n)}{(x+y)} \left(1 + \frac{dy}{dx}\right)$$

$$\Rightarrow \left(\frac{n}{y} - \frac{m+n}{x+y}\right) \frac{dy}{dx} = \frac{m+n}{x+y} - \frac{m}{x}$$

$$\Rightarrow \left(\frac{nx+ny-my-ny}{y(x+y)}\right) \frac{dy}{dx} = \left(\frac{mx+nx-mx-my}{(x+y)x}\right)$$

$$\Rightarrow \frac{nx-my}{y(x+y)} \cdot \frac{dy}{dx} = \frac{nx-my}{x(x+y)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x}$$

3. The correct option is (B) $\frac{27}{5}$

$$= 3^{3-\log_3 5}$$

$$\Rightarrow \frac{3^3}{3^{\log_3 5}}$$

$$\Rightarrow \frac{27}{5}$$

4. The correct option is (C) $\frac{1}{2} \sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}} + c$

$$I = \int \frac{x^2-1}{x^3 \sqrt{2x^4-2x^2+1}} dx = \int \frac{\left(\frac{1}{x^3} - \frac{1}{x^5}\right) dx}{\sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}}}$$

Put

$$2 - \frac{2}{x^2} + \frac{1}{x^4} = z$$

$$\Rightarrow I = \frac{1}{4} \int \frac{dz}{\sqrt{z}}$$

$$= \frac{1}{2} \times \sqrt{z} + c \Rightarrow \frac{1}{2} \sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}} + c.$$

5. The correct option is (C) $\frac{9}{2}$.

The sides of the parallelepiped are:

$$\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\vec{b} = \hat{i} + \alpha\hat{j} + 2\hat{k}$$

$$\vec{c} = \hat{i} + 2\hat{j} + \alpha\hat{k}$$

$$\therefore \text{Volume} = [\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} 2 & 3 & 4 \\ 1 & \alpha & 2 \\ 1 & 2 & \alpha \end{vmatrix} = 15$$

$$\Rightarrow 2(\alpha^2 - 4) + 3(2 - \alpha) + 4(2 - \alpha) = 15$$

$$\Rightarrow 2\alpha^2 - 8 + 6 - 3\alpha + 8 - 4\alpha = 15$$

$$\Rightarrow 2\alpha^2 - 7\alpha - 9 = 0$$

$$\Rightarrow 2\alpha^2 - 9\alpha + 2\alpha - 9 = 0$$

$$\Rightarrow \alpha(2\alpha - 9) + (2\alpha - 9) = 0$$

$$\Rightarrow (2\alpha - 9)(\alpha + 1) = 0$$

$$\Rightarrow 2\alpha - 9 = 0 \text{ OR } \alpha + 1 = 0$$

$$\Rightarrow \alpha = \frac{9}{2} \text{ OR } \alpha = -1.$$

6. The correct option is (D) 8.

$$= \frac{2c^2}{ab}$$

$$= \frac{2 \times 4}{1}$$

$$= 8$$

7. The correct option is (D) $(b-a)^2$

$$\int_a^b |x-a| + |x-b| dx \quad (0 < a < b)$$

$$\Rightarrow \int_a^b |x-a| + \int_a^b |x-b| dx$$

$$\Rightarrow \int_a^b |x-a| dx + \int_b^a |x-b| dx$$

$$\Rightarrow \int_a^b x \cdot dx - \int_a^b a \cdot dx + \int_b^a x \cdot dx - \int_b^a 1 \cdot dx$$

$$\Rightarrow \left[\frac{x^2}{2}\right]_a^b - a[x]_a^b + \left[\frac{x^2}{2}\right]_b^a - b[x]_b^a$$

$$\Rightarrow \frac{b^2 - a^2}{2} - a[b-a] + \frac{a^2 - b^2}{2} - b[a-b]$$

$$\Rightarrow -ab + a^2 - ab + b^2$$

$$\Rightarrow b^2 + a^2 - 2ab$$

$$\Rightarrow (b-a)^2$$

8. The correct option is (A) 2, -6

$$\begin{aligned} |\hat{a}\hat{b}\hat{c}| &= 7 \\ \hat{a} \cdot (\hat{b} \times \hat{c}) &= 7 \\ \hat{b} \times \hat{c} &= \begin{vmatrix} i & j & k \\ 1 & \lambda & -2 \\ 1 & 1 & 1 \end{vmatrix} \end{aligned}$$

$$\begin{aligned} \Rightarrow i(\lambda + 2) + j(1 + 2) + k(1 - \lambda) \\ \Rightarrow a \cdot (b \times c) \Rightarrow \lambda(\lambda + 2) - 3 - 2(1 - \lambda) = 7 \\ \Rightarrow \lambda^2 + 2\lambda + 2\lambda - 5 = 7 \\ \Rightarrow \lambda^2 + 4\lambda - 12 = 0 \\ \Rightarrow \lambda = 2, -6. \end{aligned}$$

9. The correct option is (C) $2m - n = 1$

$$\begin{aligned} |P(A)| &= 2^m, |P(B)| = 2^n \\ 2^m - 2^n &= 112, \\ \text{only possible values are } m &= 7, n = 4. \\ \text{and only option (c) does not satisfy it.} \end{aligned}$$

10. (C) $\frac{3}{2}$.

11. The correct option is (A) 2:1

Let A, B and C be the angles of a triangle.
Let a, b and c be the opposite sides of these angles respectively.

According to the question,

$$\tan \left[\frac{1}{2}(A - B) \right] = \frac{1}{3} \tan \left[\frac{1}{2}(A + B) \right]$$

According to Napier's Analogy, we can write the above equation as

$$\frac{a-b}{a+b} \cot \frac{C}{2} = \frac{1}{3} \tan \left[90 - \frac{C}{2} \right]$$

$$\frac{a-b}{a+b} \cot \frac{C}{2} = \frac{1}{3} \cot \frac{C}{2}$$

Cancelling $\cot \frac{C}{2}$ from the equation, we get

$$\begin{aligned} \frac{a-b}{a+b} &= \frac{1}{3} \\ 3a - 3b &= a + b \\ 2a &= 4b \\ a &= 2b \\ a:b &= 2:1 \end{aligned}$$

Hence, option (A) is the correct answer

12. The correct option is (A) $x = n\pi \pm \frac{\pi}{4}, n \in \mathbb{Z}$.

Here, we can write the equation as:

$$\begin{aligned} 4 \cos^2 x + 6 \sin^2 x &= 5 \\ \Rightarrow 4(1 - \sin^2 x) + 6 \sin^2 x &= 5 \end{aligned}$$

$$\Rightarrow 4 - 4 \sin^2 x + 6 \sin^2 x = 5$$

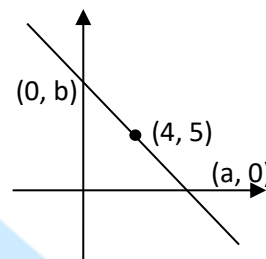
$$\Rightarrow 2 \sin^2 x = 1$$

$$\Rightarrow \sin^2 x = \frac{1}{2} = \left(\frac{1}{\sqrt{2}} \right)^2 = \sin^2 \frac{\pi}{4}$$

Thus, solution can be written as:

$$x = n\pi \pm \frac{\pi}{4}, n \in \mathbb{Z}.$$

13. The correct option is (D) $5x + 4y = 40$.



$$\Rightarrow \frac{x}{a} + \frac{y}{b} = 1 \quad \dots\dots\dots(i)$$

$$\Rightarrow \frac{a + 0}{2} = 4$$

$$\Rightarrow a = 8$$

$$\Rightarrow \frac{b + 0}{2} = 5$$

$$\Rightarrow b = 10$$

$$\text{So, } \frac{x}{8} + \frac{y}{10} = 1$$

$$\Rightarrow 5x + 4y = 40$$

14. The correct option is (A) $[-1, 0) \cup \{1\}$.

Let $g(x) = \cos^{-1}x$ and $h(x) = [x]$.

$$\text{Then, } f(x) = \frac{g(x)}{h(x)}$$

$$\therefore \text{Domain}(f) = \text{Domain}(g) \cap \text{Domain}(h) - \{x: h(x) = 0\}$$

$$\text{Clearly, Domain}(g) = [-1, 1], \text{Domain}(h) = \mathbb{R} - \{0\},$$

$$\therefore \text{Domain}(f) = [-1, 1] \cap \mathbb{R} - \{0\} = [-1, 0) \cup \{1\}.$$

15. The correct option is (C) 1

$$\tan 30^\circ + \tan 15^\circ = -p$$

$$\tan 30^\circ \times \tan 15^\circ = q$$

$$\tan 45^\circ = \frac{\tan 30^\circ + \tan 15^\circ}{1 - \tan 30^\circ \tan 15^\circ}$$

$$1 = \frac{-p}{1 - q}$$

$$1 - q = -p$$

$$1 + p - q = 0$$

$$2 + p - q = 1$$

correct answer is option (C).



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16. The correct option is (D) $\lim_{x \rightarrow 0+} \frac{\cos x}{1+2x} = 0$

Only option (D) is solved wrong.

17. The correct option is (B) $\left(\frac{1}{2}\right) \log_{10} \left(\frac{x+1}{x-1}\right)$

Explanation for the correct option:

Compute the required value:

$$\begin{aligned} y &= \frac{10^x - 10^{-x}}{10^x + 10^{-x}} \\ \Rightarrow y &= \frac{10^{2x} - 1}{10^{2x} + 1} \\ \Rightarrow \frac{y+1}{y-1} &= 10^{2x} \\ \Rightarrow \log \left(\frac{y+1}{y-1} \right) &= 2x \log(10) \\ \Rightarrow \frac{1}{2} \log_{10} \left(\frac{y+1}{y-1} \right) &= x \\ \therefore f^{-1}(x) &= \left(\frac{1}{2} \right) \log_{10} \left(\frac{x+1}{x-1} \right) \end{aligned}$$

Hence, the correct option is Option (B)

18. The correct option is (C) (3, 15).

19. The correct option is (A) $\cot a_1 - \cot a_n$

Explanation for the correct option:

Find the value of

$$\begin{aligned} &(\sin d)[\operatorname{cosec} a_1 \operatorname{cosec} a_2 + \operatorname{cosec} a_2 \operatorname{cosec} a_3 + \dots + \operatorname{cosec} a_{(n-1)} \operatorname{cosec} a_n] : \\ &\text{Given } a_1, a_2, \dots, a_n \text{ are in AP.} \\ &\text{Common difference, } d = a_2 - a_1 \\ &= a_3 - a_2 = a_n - a_{(n-1)} \\ &(\sin d)[\operatorname{cosec} a_1 \operatorname{cosec} a_2 + \operatorname{cosec} a_2 \operatorname{cosec} a_3 + \dots + \operatorname{cosec} a_{(n-1)} \operatorname{cosec} a_n] \\ &= \frac{\sin(a_2 - a_1)}{\sin a_1 \sin a_2} + \frac{\sin(a_3 - a_2)}{\sin a_2 \sin a_3} + \dots + \frac{\sin(a_n - a_{(n-1)})}{\sin a_{(n-1)} \sin a_n} \\ &= \frac{(\sin a_2 \cos a_1 - \sin a_1 \cos a_2)}{\sin a_1 \sin a_2} + \frac{(\sin a_3 \cos a_2 - \sin a_2 \cos a_3)}{\sin a_2 \sin a_3} \\ &+ \dots + \frac{(\sin a_n \cos a_{(n-1)} - \sin a_{(n-1)} \cos a_n)}{\sin a_{(n-1)} \sin a_n} \\ &[\because \sin(A - B) = \sin A \cos B - \cos A \sin B] \\ &= (\cot a_1 - \cot a_2) + (\cot a_2 - \cot a_3) + \dots \\ &\quad + (\cot a_{(n-1)} - \cot a_n) \\ &= (\cot a_1 - \cot a_n) \end{aligned}$$

Hence, the correct option is A.

20. The correct option is (C) 0, -1

$$\tan^{-1} \sqrt{x(x+1)} + \sin^{-1} \sqrt{x^2 + x + 1} = \frac{\pi}{2}$$

$$\tan y = \sqrt{x(x+1)}$$

$$\cos y = \frac{1}{\sqrt{x^2 + x + 1}}$$

$$y = \cos^{-1} \left(\frac{1}{\sqrt{x^2 + x + 1}} \right)$$

$$\cos^{-1} \left(\frac{1}{\sqrt{x^2 + x + 1}} \right) + \sin^{-1} (\sqrt{x^2 + x + 1}) = \frac{\pi}{2}$$

We know that

$$\sin^{-1} y + \cos^{-1} y = \frac{\pi}{2}$$

$$\text{So } \frac{1}{\sqrt{x^2 + x + 1}} = \sqrt{x^2 + x + 1}$$

$$x^2 + x + 1 = 1$$

$$x^2 + x = 0$$

$$x = 0, -1$$

21. The correct option is (D) $\frac{5}{4}$

$$\operatorname{cosec} \theta - \cot \theta = 2 \quad \dots (1)$$

by using property

$$\operatorname{cosec} \theta + \cot \theta = \frac{1}{2} \quad \dots (2)$$

Simultaneously solving ... (1) and ... (2), we get

$$\operatorname{cosec} \theta = \frac{5}{4}$$

correct answer is option (D).

22. The correct option is (A) Differentiable at $x = 0$.

$$f(x) = (1 + 2x)^{\frac{1}{x}}$$

$$f(x) = \lim_{x \rightarrow 0} (1 + 2x)^{\frac{1}{x}}$$

$$= e^{\lim_{x \rightarrow 0} \frac{2x}{x}}$$

$$= e^2$$

$$L.H.S. = R.H.S. = f(0)$$

Hence, the given function is continuous.

for differentiability at $(x = 0)$

$$= e^{\lim_{x \rightarrow 0} \frac{2x}{x}}$$

$$= e^2$$

Hence, the given function is differentiable at $x = 0$.

23. The correct option is (B) 16.

24. The correct option is (C) $\vec{a}$ and $\vec{c}$ are collinear

$$(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{c} \cdot \vec{a}) \cdot \vec{b} - (\vec{c} \cdot \vec{b}) \cdot \vec{a} \quad \dots (1)$$

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \cdot \vec{b} - (\vec{a} \cdot \vec{b}) \cdot \vec{c} \quad \dots (2)$$

$$(\vec{c} \cdot \vec{a}) \cdot \vec{b} - (\vec{c} \cdot \vec{b}) \cdot \vec{a} = (\vec{a} \cdot \vec{c}) \cdot \vec{b} - (\vec{a} \cdot \vec{b}) \cdot \vec{c}$$



$$(\vec{c} \cdot \vec{b}) \cdot \vec{a} = (\vec{a} \cdot \vec{b}) \cdot \vec{c}$$

$$\vec{a} = \frac{(\vec{a} \cdot \vec{b})}{(\vec{c} \cdot \vec{b})} \vec{c}$$

$\vec{a}$ and $\vec{c}$ are collinear option ... (C).

25. The correct option is (A) 125

If $\lambda = 5$, then $y = -\frac{x}{2}$.

This implies $x^2 + \left(-\frac{x}{2}\right)^2 = 25$ or equivalently $x = 2\sqrt{5}$

and $y = -\frac{x}{2} = -\sqrt{5}$ or $x = -2\sqrt{5}$ and $y = -\frac{x}{2} = +\sqrt{5}$.

The temperature at the points $(2\sqrt{5}, -\sqrt{5})$ and $(-2\sqrt{5}, \sqrt{5})$ are maximum values given by

$$\begin{aligned} &4x^2 - 4xy + y^2 \\ &= 4(2\sqrt{5})^2 - 4(2\sqrt{5})(-\sqrt{5}) + (-\sqrt{5})^2 \\ &= 4 \cdot 4 \cdot 5 + 8 \cdot 5 + 5 \\ &= 80 + 40 + 5 \\ &= 125 \end{aligned}$$

26. The correct option is (B) an odd function.

$$f(x) = \log(x + \sqrt{x^2 + 1})$$

$$\Rightarrow f(-x) = \log(-x + 1)$$

$$f(x) + f(-x) = \log(x + \sqrt{x^2 + 1}) + \log(-x + \sqrt{x^2 + 1})$$

$$f(x) + f(-x) = \log[(x + \sqrt{x^2 + 1})(-x + \sqrt{x^2 + 1})]$$

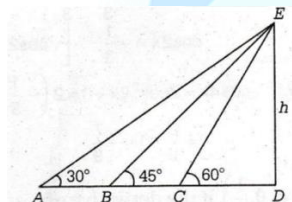
$$f(x) + f(-x) = \log(x^2 + 1 - x^2)$$

$$f(x) + f(-x) = 0$$

$$f(-x) = -f(x)$$

Hence $f(x)$ is odd function.

27. The correct option is (A) $\sqrt{3} : 1$



In $\triangle EDA$,

$$\tan 30^\circ = \frac{ED}{AD}$$

$$\frac{1}{\sqrt{3}} = \frac{ED}{AD} = \frac{h}{AD}$$

$$\Rightarrow AD = h\sqrt{3}$$

In $\triangle EDB$,

$$\tan 45^\circ = \frac{h}{BD} \Rightarrow BD = h$$

In $\triangle EDC$,

$$\tan 60^\circ = \frac{h}{CD} \Rightarrow CD = \frac{h}{\sqrt{3}}$$

Now,

$$\frac{AB}{BC} = \frac{AD-BD}{BD-CD} \Rightarrow \frac{AB}{BC} = \frac{h\sqrt{3}-h}{h-\frac{h}{\sqrt{3}}}$$

$$\Rightarrow \frac{AB}{BC} = \frac{AD-BD}{BD-CD}$$

$$\Rightarrow \frac{AB}{BC} = \frac{\sqrt{3}-1}{(\sqrt{3}-1)} \times \sqrt{3}$$

$$\Rightarrow \frac{AB}{BC} = \frac{\sqrt{3}}{1}$$

$$\therefore AB:BC = \sqrt{3}:1$$

28. The correct option is (C) 38

$$\text{Mean} = \frac{\text{sum of observation}}{\text{number of observation}}$$

$$38 = \frac{\text{sum of observation}}{25}$$

$$950 = \text{sum of observation}$$

$$\text{ACT} = 950 - (23 + 38) + (25 + 36) = 950$$

The sum of observation remains the same so the mean will be 38.

29. The correct option is (B) 1

Let a be the first term and d be the common difference of the corresponding AP. Then

$$p^{\text{th}} \text{ term of AP } (T_p) = a + (p-1)d = 1/p \quad \dots(1)$$

$$q^{\text{th}} \text{ term of AP } (T_q) = a + (q-1)d = 1/p \quad \dots(2)$$

From Eqs. (1) and (2),

$$(p-q)d = \frac{1}{q} - \frac{1}{p} = \frac{p-q}{pq} \Rightarrow d = \frac{1}{pq}$$

Form Eq. (1),

$$a + (p-1)\frac{1}{pq} = \frac{1}{q} \Rightarrow a = \frac{1}{pq}$$

Therefore,

$$\begin{aligned} T_{pq} &= a + (pq-1)d \\ &= \frac{1}{pq} + (pq-1)\frac{1}{pq} = 1 \end{aligned}$$

So, pq^{th} term is 1.

30. The correct option is (C) $\frac{1}{5}$.

$$\begin{aligned} \text{Total cases} &= 5 \times 4 \times 3 \times 2 \\ &= 120 \end{aligned}$$

no. divisible by 3 can only be formed by given number if we select 1, 2, 4, 5 and for arranging them



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we have 4! Ways

$$\text{Probability} = \frac{24}{120} = \frac{1}{5}$$

31. The correct option is (A) $\vec{b} = 5\hat{i} + \hat{j} + 2\hat{k}$.

$$\text{Let } \vec{b} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{a}\vec{b} = 2x + 2y + z = 14 \quad \dots(1)$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 1 \\ x & y & z \end{vmatrix} = (2z - y)\hat{i} - (2z - x)\hat{j} +$$

$$(2y - 2x)\hat{k}$$

$$\Rightarrow (2z - y)\hat{i} - (2z - x)\hat{j} + (2y - 2x)\hat{k} = 3\hat{i} + \hat{j} - 8\hat{k}$$

$$2z - y = 3 \quad \dots(2)$$

$$2z - x = -1 \quad \dots(3)$$

$$y - x = -4 \quad \dots(4)$$

Solving (1), (2), (3), (4), we get

$$x = 5, y = 1, z = 2$$

$$\Rightarrow \vec{b} = 5\hat{i} + \hat{j} + 2\hat{k}$$

32. The correct option is (A) $n - m = 56$.

33. The correct option is (B) $8x^2 + 9y^2 = 72$

If we go with options, option (B) $8x^2 + 9y^2 = 72$

$$\frac{x^2}{9} + \frac{y^2}{8} = 1$$

$$\text{for e, } e = \sqrt{1 - \frac{b^2}{a^2}}$$

$$= \sqrt{1 - \frac{8}{9}}$$

$$= \pm \frac{1}{3}$$

hence option (B) is the answer, as it satisfies the condition.

34. The correct options are B) and D)

Converting infinite series into definite integral

$$\text{i.e. } \lim_{n \rightarrow \infty} \frac{h(n)}{n}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=g(n)}^{k(n)} f\left(\frac{r}{n}\right) = \int f(x) dx$$

$$\lim_{n \rightarrow \infty} \frac{g(n)}{n}, \text{ where } \frac{r}{n} \text{ is replaced with } x.$$

$\sum$ is replaced with integral

$$\text{Here, } \lim_{n \rightarrow \infty} \frac{1^n + 2^n + \dots + n^n}{(n+1)^{n-1} \{(n+1) + (n+2) + \dots + n\}} = \frac{1}{60}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n r^n}{(n+1)^{n-1} \left[n^2 a + \frac{n(n+1)}{2} \right]} = \frac{1}{60}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{2 \sum_{r=1}^n \left(\frac{r}{n}\right)^n}{\left(1 + \frac{1}{n}\right)^{n-1} (2na + n + 1)} = \frac{1}{60}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n} \left(2 \sum_{r=1}^n \left(\frac{r}{n}\right)^n \right) \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^{n-1} (2a+1)} = \frac{1}{60}$$

$$\Rightarrow 2 \int_0^1 (x^4) dx \cdot \frac{1}{1 \cdot (2a+1)} = \frac{1}{60}$$

$$\Rightarrow \frac{2 \left[\frac{x^{4+1}}{4+1} \right]_0^1}{(2a+1)(a+1)} = \frac{1}{60}$$

$$\therefore \frac{2}{(2a+1)(a+1)} = \frac{1}{60}$$

$$\Rightarrow (2a+1)(a+1) = 120$$

$$\Rightarrow 2a^2 + 3a - 120 = 0$$

$$\Rightarrow 2a^2 + 3a - 119 = 0$$

$$\Rightarrow (2a+17)(a-7) = 0$$

$$\Rightarrow a = 7, \frac{-17}{2}$$

35. The correct option is (A) $40b = 11\sqrt{5}a$.

For a, we have lines,

$$-2x + y = 2$$

$$2x - y = 2$$

For b, we have lines,

$$4x - 3y = 5$$

$$4x - 3y = -\frac{1}{2}$$

using distance formula between parallel lines =

$$\left| \frac{c_1 - c_2}{\sqrt{a^2 + b^2}} \right|$$

we get, $a = \frac{4}{\sqrt{5}}$ and $b = \frac{11}{10}$, these both values satisfy in option (A) only, so the answer is option (A).

36. The correct option is (C) Divisible by $(1+x)$ and $(1+y)$.

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 5 \end{vmatrix}$$

$$= (20 - 1) - 1(5 - 1) + 1(1 - 4)$$

$$= 12$$

So, option (C) is correct.

37. The correct option is (C)

Given,

$$P(A \cup B) = P(A) + P(B) - P(A)P(B)$$

$$1) P\left(\frac{B}{A}\right) = P\left(\frac{P(B \cap A)}{A}\right) = P\left(\frac{P(B) \times P(A)}{A}\right) = P(B)$$

$$2) P(A^c \cup B^c) = P(A^c) + P(B^c) - P(A^c \cap B^c)$$

$$3) P((A \cup B)^c) = (1 - P(A \cup B)) = (1 - P(A) - P(B) + P(A)P(B))$$

$$\Rightarrow P((A \cup B)^c) = (1 - P(A))(1 - P(B)) = P(A^c)P(B^c).$$





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$$4) P\left(\frac{A}{B}\right) = \left(\frac{P(A \cap B)}{B}\right)$$

$$P\left(\frac{A}{B}\right) = \left(\frac{P(A)P(B)}{B}\right) = P(A)$$

Therefore, Option C is correct

$$\therefore h = -\frac{7}{2}, k = \frac{5}{2}, a = -\frac{3}{4}$$

$$\text{Vertex } A \text{ is } \left(-\frac{7}{2}, \frac{5}{2}\right)$$

$$\text{Focus is } s(h + a, k) = \left(-\frac{7}{2} - \frac{3}{4}, \frac{5}{2}\right) = \left(-\frac{17}{4}, \frac{5}{2}\right)$$

38. The correct option is (B)

Vector $a\hat{i} + a\hat{j} + c\hat{k}$, $\hat{i} + \hat{k}$ and $c\hat{i} + c\hat{j} + b\hat{k}$, are coplanar

$$\therefore \begin{vmatrix} a & a & c \\ 1 & 1 & 0 \\ c & c & b \end{vmatrix} = 0$$

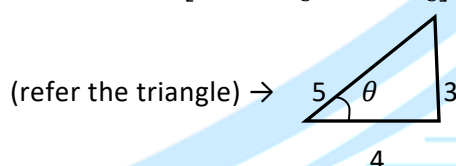
$$\Rightarrow -1(ab - c^2) - 1(ac - ac) = 0$$

$$\Rightarrow c^2 = ab$$

$$\Rightarrow a, b, c \text{ are in G.P.}$$

39. The correct option is (B)

$$\text{Given: } \cot \left[\operatorname{cosec}^{-1} \frac{5}{3} + \tan^{-1} \frac{2}{3} \right]$$



$$\therefore \operatorname{cosec}^{-1} \frac{5}{3} = \tan^{-1} \frac{3}{4}$$

$$\therefore \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{2}{3} = \tan^{-1} \left[\frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{3}{4} \times \frac{2}{3}} \right]$$

$$= \tan^{-1} \left[\frac{17}{6} \right] = \cot^{-1} \left[\frac{6}{17} \right]$$

$$\therefore \cot \left[\cot^{-1} \left(\frac{6}{17} \right) \right] = \frac{6}{17}$$

Option 'A' is correct.

40. The correct option is (B) 11.

$$\text{Given series is } \frac{\sqrt{5}}{3}, \frac{\sqrt{5}}{4}, \frac{\sqrt{5}}{5}, \dots$$

$$\sqrt{5} \left(\frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots \right)$$

this is a H.P. and 13 will be its n^{th} term

$$13 = 3 + (n - 1)d$$

$$11 = n$$

$$\frac{\sqrt{5}}{13} \text{ is its } 11^{\text{th}} \text{ term}$$

correct answer is option (B).

41. The correct option is (C)

$$4y^2 - 20y = -12x - 67$$

$$y^2 - 5y = -3x - \frac{67}{4}$$

$$\Rightarrow \left(y - \frac{5}{2}\right)^2 - \frac{25}{4} = -3x - \frac{67}{4}$$

$$\Rightarrow \left(y - \frac{5}{2}\right)^2 = -3x - \frac{42}{4} = -3\left(x + \frac{7}{2}\right)$$

$$\Rightarrow \left(y - \frac{5}{2}\right)^2 = -3\left[x - \left(-\frac{7}{2}\right)\right]$$

42. The correct option is (B) 23, 32.

Observations in ascending order 23, 29, 32, 49, 118.

$$n = 5$$

$$\therefore p^{\text{th}} \text{ Percentile} = \left(\frac{p}{100}\right) \times n.$$

$$10^{\text{th}} \text{ percentile} = \frac{10}{100} \times 5 = 0.5 \text{ ie. } 1^{\text{st}} \text{ data} = 23$$

$$50^{\text{th}} \text{ percentile} = \frac{50}{100} \times 5 = 2.5 \text{ ie. } 3^{\text{rd}} \text{ data} = 32.$$

43. The correct option is (A) 1/5.

$$4x - y = 0 \quad \dots (1)$$

$$4x - y + 1 = 0 \quad \dots (2)$$

$$x + y = 0 \quad \dots (3)$$

$$x + y - 1 = 0 \quad \dots (4)$$

$$\text{Area of parallelogram} = \left| \frac{(c_1 - c_2)(d_1 - d_2)}{ab' - ba'} \right|$$

$$\text{from } \dots (1) a = 4, b = -1, c_1 = 0$$

$$\text{from } \dots (2) a = 4, b = -1, c = 1$$

$$\text{from } \dots (3) a = 1, b = 1, d_1 = 0$$

$$\text{from } \dots (4) a = 1, b = 1, d_2 = -1$$

putting values in formula, we get

$$= \left| \frac{1}{-1-4} \right| = \frac{1}{5}$$

44. The correct option is (B) $n \sum_{i=1}^n a_i^2 \geq (\sum_{i=1}^n a_i)^2$.

Variance ≥ 0

$$n \sum_{i=1}^n a_i^2 \geq (\sum_{i=1}^n a_i)^2$$

45. The correct option is (D) 1/4.

Circle C1: $x^2 + y^2 - 2y = 0$ can be rewritten as C1:

$$(x - 0)^2 + (y - 1)^2 = 1^2$$

Circle C2: $x^2 + y^2 - 2y = 0$ can be rewritten as C2:

$$(x - 0)^2 + (y - 1)^2 = 2^2$$

Hence, C1 and C2 are concentric circles.

Required point belongs to the area occupied by

bigger circle (A_2) and it is excluded in the region of

area of similar circle (A_1).

$$\text{Hence, required probability} = \frac{A_2 - A_1}{A_2} = \frac{\pi \cdot 2^2 - \pi \cdot 1^2}{\pi \cdot 2^2} =$$

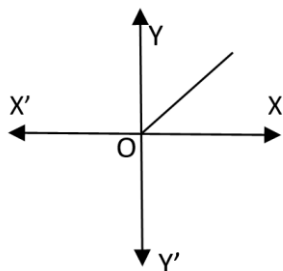
$$\frac{3\pi}{4\pi} = \frac{3}{4}$$

46. The correct option is (A) $(-\infty, \infty)$.

$$\text{Given, } f(x) = x + |x|$$



$$\therefore f(x) = \begin{cases} 2x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$



It is clear from the graph of $f(x)$ is continuous for every value of x

47. The correct option is (B) $\pi - \cos^{-1}(x)$.

$$\cos^{-1}(-x) = \pi - \cos^{-1}(x)$$

$$\text{Let } \cos^{-1}(-x) = y \text{ i.e., } -x = \cos y$$

Thus,

$$\cos^{-1}(x) = \pi - y$$

$$\text{Or, } \cos^{-1}(x) = \pi - \cos^{-1}(-x)$$

$$\text{Therefore, } \cos^{-1}(-x) = \pi - \cos^{-1}(x)$$

48. The correct option is (B) $36 \sin^2 \theta$

We know

$$\cos^{-1}x + \cos^{-1}y$$

$$= \cos^{-1}[xy - \sqrt{1-x^2}\sqrt{1-y^2}]$$

Now,

$$\cos^{-1}\frac{x}{2} + \cos^{-1}\frac{y}{3} = \theta$$

$$\Rightarrow \cos^{-1}\left[\frac{xy}{2 \cdot 3} - \sqrt{1-\frac{x^2}{4}}\sqrt{1-\frac{y^2}{9}}\right] = \theta$$

$$\Rightarrow \frac{xy}{2 \cdot 3} - \sqrt{1-\frac{x^2}{4}}\sqrt{1-\frac{y^2}{9}} = \cos \theta$$

$$\Rightarrow xy - \sqrt{4-x^2}\sqrt{9-y^2} = 6 \cos \theta$$

$$\Rightarrow \sqrt{4-x^2}\sqrt{9-y^2} = xy - 6 \cos \theta$$

$$\Rightarrow (4-x^2)(9-y^2) = x^2y^2 + 36\cos^2\theta - 12xy \cos \theta$$

$$\Rightarrow 36 - 4y^2 - 9x^2 + x^2y^2 = x^2y^2 + 36\cos^2\theta - 12xy \cos \theta$$

$$\Rightarrow 36 - 4y^2 - 9x^2 = 36\cos^2\theta - 12xy \cos \theta$$

$$\Rightarrow 9x^2 - 12xy \cos \theta + 4y^2 = 36 - 36\cos^2\theta$$

$$\Rightarrow 9x^2 - 12xy \cos \theta + 4y^2 = 36\sin^2\theta$$

49. The correct option is (C)

$$a^2 - b^2 = 2c^2$$

$$\text{Given } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

...(1)

On differentiating w.r.t. x , we get

$$\begin{aligned} \frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} &= 0 \\ \Rightarrow \frac{dy}{dx} &= -\frac{xb^2}{a^2y} \end{aligned}$$

$$\text{and } x^2 - y^2 = c^2$$

On differentiating w.r.t. x , we get

$$\begin{aligned} 2x - 2y \frac{dy}{dx} &= 0 \\ \Rightarrow \frac{dy}{dx} &= \frac{x}{y} \end{aligned}$$

The two curves will cut at right angles, if

$$\begin{aligned} \left(\frac{dy}{dx}\right)_{c1} \times \left(\frac{dy}{dx}\right)_{c2} &= -1 \\ \Rightarrow -\frac{b^2x}{a^2y} \cdot \frac{x}{y} &= -1 \\ \Rightarrow \frac{x^2}{a^2} &= \frac{y^2}{b^2} \\ \Rightarrow \frac{x^2}{a^2} &= \frac{y^2}{b^2} = \frac{1}{2} \end{aligned}$$

[using eq. (i)]

On substituting these values in $x^2 - y^2 = c^2$, we get,

$$\begin{aligned} \frac{a^2}{2} - \frac{b^2}{2} &= c^2 \\ \Rightarrow a^2 - b^2 &= 2c^2 \end{aligned}$$

Hence, option C is correct.

50. The correct option is (B) $\frac{1}{2} \times A_{n+1}$.

Let α, β be the roots of $x^2 - x - 1 = 0$.

Then, $\alpha + \beta = 1$, $\alpha\beta = -1$ or, $\frac{1}{\alpha} + \frac{1}{\beta} = -1$... (1)

Now, $A_n = \alpha^n + \beta^n$, $A_{n+1} = \alpha^{n+1} + \beta^{n+1}$,

$$A_{n-1} = \alpha^{n-1} + \beta^{n-1}$$

The A.M. of A_{n-1} and A_n is,

$$\begin{aligned} \frac{A_{n-1} + A_n}{2} &= \frac{\alpha^{n-1} + \beta^{n-1} + \alpha^n + \beta^n}{2} \\ &= \frac{\alpha^{n-1}(\alpha + 1) + \beta^{n-1}(\beta + 1)}{2} \\ &= \frac{\alpha^{n-1}(-\frac{1}{\beta}) + \beta^{n-1}(-\frac{1}{\alpha})}{2} \end{aligned}$$

[Form (1)]

$$= -\frac{\alpha^{n+1} + \beta^{n+1}}{2\alpha\beta}$$

$$= -\frac{\alpha^{n+1} + \beta^{n+1}}{2(-1)} = \frac{\alpha^{n+1} + \beta^{n+1}}{2}$$

$$= \frac{1}{2} \times A_{n+1}$$



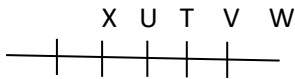
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Reasoning:

51. (D) Between U and V

Explanation:



52. (A) JWW.

D E F G H I J K L
N O P Q R S T U V W X Y Z
N O P Q R S T U V W X Y Z

53. (A) **Catastrophe** - a sudden disaster that causes great suffering or damage.

Debacle - an event or situation that is a complete failure and causes embarrassment.

54. (B) PMK

HEC, JGE, LIG, NKI, _____
H I J K L M N O P
E F G H I J K L M
C D E F G H I J K

55. (C) D

Paint - Brush
 Thread - Needle

56. (A) **Driving**

Explanation: Driving is done on the road. While, Diving, Swimming and Sailing is done in the water.

57. (D) **Predecessor**- The person who was in the job or position before the person who is in it now

Frugal- The one who spent required amount

Extravagant - The one who spent too much money

Precursor - a person or thing that comes before somebody/something similar, especially something that develops from it or is influenced by it

58. (B) 81

Explanation: $6 + 5 = 11$

$$11 + 10 = 21$$

$$21 + 15 = 36$$

$$36 + 20 = 56$$

$$56 + 25 = 81$$

So, 81 is the Answer.

59. (A) His son.

60. (D) Horse : Speed

Butterfly and freedom are inter-related as well as, Horse and Speed are similar.

61. (C) NVNCBJ

Explanation:

M - N

U - V

M - N

B - C

A - B

I - J

Next letters of every letters decode the word.

So, the answer is NVNCBJ.

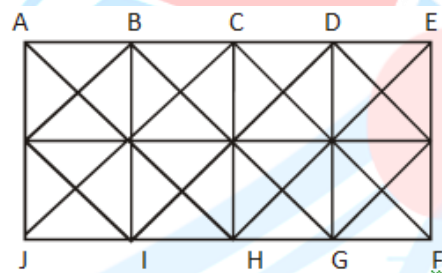
62. (C)

Explanation: Liquid melts, same Solid freezes

So, SOLID is the answer.

63. (D) 24

The figure may be labeled as shown.



The squares composed of two components each are BNQM, CORN, DPSO, MQTL, NRUQ, OSVR, PFWS, QUJT, RVIU and SWHV i.e. 10 in number.

The squares composed of four components each are ABQL, BCRQ, CDSR, DEFS, LQJK, QRIJ, RSHI and SFGH i.e. 8 in number.

The squares composed of eight components each are BRJL, CSIQ and DFHR i.e. 3 in number.

The squares composed of sixteen components





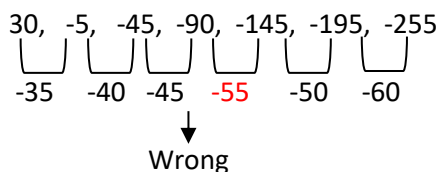
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74. (C) -145



75. (D) E and F

Books	Covers	Volume	Law/Medical
A	Yellow	New	Law
B	Green	New	Law
C	Green	Old	Law
D	Yellow	New	Medical
E	Green	Old	Medical
F	Green	Old	Medical

Thus, E and F are old medical Books and have green covers.

76. (C) 1026

$$(3800 \times 15\%) + (3800 \times 12\%) = 1026$$

77. (C) 1018

$$(3800 \times 8\%) + (4200 \times 17\%) = 1018$$

78. (A) 19 : 27

$$(38001 \times 14\%) = 532$$

$$(4200 \times 18\%) = 756$$

$$\begin{aligned} \text{Required ratio} &= 532 : 756 \\ &= 19 : 27 \end{aligned}$$

79. (B) All I, II and III follows.

Explanation: All causes of action are required.

80. (A) Only I and III follows

Explanation: I and II are required action, but instead of raising the problem to UNESCO they should raise it to WHO.

Solutions No. (81 to 85):

1 st	B
2 nd	D
3 rd	Break
4 th	C
5 th	A
6 th	E

81. (C) 3rd

82. (A) B

83. (D) None of these

84. (C) C

85. (C) None of these

86. (A) 127

$$(1657 - 6) = 1651 \div 127 = 13$$

$$(2037 - 5) = 2032 \div 127 = 16$$

87. (D) $16\frac{4}{11}$

$$\Rightarrow \theta = 30 \times 3 - \frac{11}{2}m$$

$$\Rightarrow 0^\circ = 90 - \frac{11}{2}m$$

$$\Rightarrow 90 = \frac{11}{2}m$$

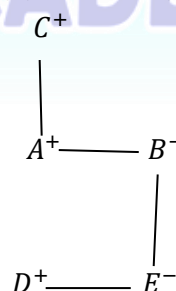
$$\Rightarrow 180 = 11m$$

$$\Rightarrow \frac{180}{11} = m$$

$$\Rightarrow 16\frac{4}{11}$$

88. (B) A

Where (+) = Female, (-) = Male





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89. (D) 1800

$$\text{Rate of machine} = \frac{270}{6}$$

No. of machines = 10

the rate of bottle producing by machine =
 45×10

= 450 per minute.

No. of bottles produced by 10 machines in 4 min.

= rate $\times$ time = work

$$450 \times 4 = 1800 \text{ bottles.}$$

90. (A) 40

Let the age of the person's mother be 'a' units.

So, age of the person = $\left(\frac{2}{5}\right)a$

after 8 years, age of the person's mother = 'a+8' units

So, age of the person after 8 years = $\frac{(a+8)}{2}$

According to question

$$\Rightarrow \left(\frac{2}{5}\right)a + 8 = \frac{(a+8)}{2}$$

$$\Rightarrow \frac{2a+40}{5} = \frac{a+8}{2}$$

$$\Rightarrow 4a + 80 = 5a + 40$$

$$\Rightarrow a = 40$$

So, the present age of his mother is 40 years.

Hence, '40 years is the correct answer.

91. (C) D

n bits binary number require d-decimal digits

$$10^d > 2^n$$

taking log both sides

$$\log_{10} 10^d > \log_{10} 2^n$$

$$d > n \log_{10} 2$$

92. (D) Real Numbers.

93. (A) XY

$$\Rightarrow (X + Y). (X + \bar{Y}). (\bar{X} + Y)$$

$$\Rightarrow [X + (Y. \bar{Y})]. (\bar{X} + Y)$$

$$\Rightarrow (X + 0)(\bar{X} + Y) \quad [\because Y. \bar{Y} = 0]$$

$$\Rightarrow X. \bar{X} + XY$$

$$\Rightarrow XY$$

$$[X. \bar{X} = 0]$$

94. (A) B.

$$\Rightarrow AB + AB' + A'C + AC$$

$$\Rightarrow A(B + B') + (A' + A)C$$

$$\Rightarrow A + C$$

95. (A) A + C

$$= (A + C)(AD + AD') + AC + C$$

$$= (A + C). A. (D + D') + (A + 1). C$$

$$= (A + C). A. 1 + C \quad [\because D + D' = 1 \text{ and } (A + 1) = 1]$$

$$= A. A + C. A + C$$

$$= A + (A + 1)C \quad [\because (A + 1) = 1]$$

$$= A + 1. C$$

$$= A + C$$

96. (B) 4.00×10^{-10} sec

Explanation:

$$\begin{aligned} \text{Clock period} &= \frac{1}{\text{clock cycle}} \\ &= \frac{1}{2500 \times 10^6} \\ &= 4 \times 10^{-10} \end{aligned}$$

97. (B) 0

$$\Rightarrow AB + A\bar{B}C + A\bar{B}\bar{C}$$

$$\Rightarrow AB + A\bar{B}(C + \bar{C})$$

$$\Rightarrow AB + A\bar{B}$$

$$\Rightarrow A(B + \bar{B})$$

$$\Rightarrow A$$

So, there is no need of NAND gate to implement the given expression.

98. (D) 64K

FFFF will be the last memory location in a memory size 64K.

The equivalent addressing ranges are 0 to 65535 in decimal and 0 to FFFF in hex. Hexadecimal (also





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base 16) is a positional numeral system with a radix, or base, of 16. It uses sixteen distinct symbols, most often the symbols 0–9 to represent values zero to nine, and A–F (or alternatively a–f) to represent values ten to fifteen.

99. (C) –32768 and 32767

100. (C) NAND.

101. (B) Factual.

102. (D) is attending.

103. (D) an, the.

an M.A. → Abbreviation.

the university.

He is an M.A. with PhD and teaches in the university.

104. (D) path paved

Tense, Path → object.

because Paved

v₂ form

Paved wrong.

105. (B) to accept something that is difficult or unpleasant.

106. (C) Bohemian.

107. (B) has been guiding

108. (A) Vertigo is a major feature of all AI applications.

109. (C) To avoid doing something.

110. (C) Curtail.

111. (B) addicted to smoking.
Smoke is an object, Smoking is a verb.

112. (A) He said that he was glad to be there that evening.

113. (B) Generous.

114. (D) Complimentary

115. (D) Peruse

116. (C) Timid

117. (A) Campus always use 'on' as a preposition.
usually but it is also accepted sometime.

118. (A) to

119. (A) No error.

120. (A) am.

